

Dataset and Model-Variable Reference

Table 1 summarises the raw data sources. Table 2 traces each of the ten model inputs from its raw file column through preprocessing to the MILP variable symbol. Columns present in the combined CSV but not consumed by the model (FCR-N prices, **Total** auction rows, **freq_avg_hz**, FCR-N activation fractions, all weather variables, and **spot_exkl_vat_ore_kwh**) are omitted.

Table 1: Raw data sources consumed by the model.

Dataset	Source / URL	Period (UTC)	Native resolution	Raw DK2 rows
Nordic system frequency	https://data.fingrid.fi/en/datasets/339 (Fingrid, dataset id 339)	2025, full year	0.1 s (10 Hz)	≈ 315 M
Day-ahead spot prices (hourly)	https://www.energidataservice.dk/tso-electricity/Elspotprices	Jan–Sep 2025	1 h	6 551
Day-ahead spot prices (15 min)	https://energidataservice.dk/tso-electricity/DayAheadPrices	Oct–Dec 2025	15 min	8 836
EC smart-meter data (<i>b</i> -type)	Enyday.com — <code>data_in/data_ec/b_data.csv</code>	2025, full year	Irregular cumulative	496 121
EC smart-meter data (<i>s</i> -type)	Enyday.com — <code>data_in/data_ec/s_data.csv</code>	2025, full year	Irregular cumulative	547 852
FCR-D / FCR-N auction prices	https://www.energidataservice.dk/tso-electricity/FcrNdDK2	2025, full year	1 h	78 840

Table 2: Model input lineage: raw column $\rightarrow$ preprocessing $\rightarrow$ model variable. In `01_Back.test.ipynb` cell 8, all **ore/kWh** columns are divided by 100 to obtain DKK/kWh before entering the solver.

Source file	Raw column(s)	Preprocessing applied	Final CSV column	Symbol	Unit
EC smart-meter data (<code>b_data.csv</code> / <code>s_data.csv</code> , Enyday.com)					
<code>b_data.csv</code>	Consumption	(1) Expand to 8760-hour grid per serial. (2)	consumption	$D_{e,t}$	kWh/h
<code>s_data.csv</code>	(cumulative kWh; load serials: 1–57 excl. 15 for b; 58–120 excl. 107 for s)	Linear-interpolate missing cumulative readings (3 299 corrected for b; 4 123 for s). (3) Hourly delta via <code>.diff()</code> . (4) Sum over all non-PV serials per hour.			

Table 2 continued...

Source file	Raw column(s)	Preprocessing applied	Final CSV column	Symbol	Unit
b_data.csv s_data.csv	Consumption (cumulative kWh; PV serial only: –1 for b-type; –2 for s-type)	Same steps (1)–(3) as above, applied to PV serial only.	pv_production_kwh	$PV_{e,t}$	kWh/h
Spot prices (Elspotprices.csv Jan–Sep; DayAheadPrices.csv Oct–Dec, Energidataservice)					
Elspot- prices.csv DayAhead- Prices.csv	SpotPriceDKK (DKK/MWh; hourly) DayAheadPrice- DKK (DKK/MWh; 15-min, averaged to hourly)	(1) Average 15-min prices within each hour (DayAheadPrices only). (2) Merge: Elspotprices takes priority for overlapping hours near the Oct transition; one missing terminal hour (2025-12-31 23:00) filled by forward-fill. (3) Convert DKK/MWh $\div 10 = \text{øre/kWh}$. (4) Add time-of-use DSO tariff (Radius 2026 schedule), transmission fee (5.80), system fee (5.40), electricity duty (0.80) [øre/kWh excl. VAT]. (5) Multiply total by 1.25 (25 % VAT).	buy_price_inkl_ vat_ore_kwh	λ_t^{im}	øre/kWh
Elspot- prices.csv DayAhead- Prices.csv	Same as above	Steps (1)–(3) as above. Subtract feed-in handling fee 0.59 øre/kWh (incl. VAT). <i>Negative values possible during negative-price hours.</i>	sell_price_inkl_ vat_ore_kwh	λ_t^{ex}	øre/kWh
FCR-D auction clearing prices (FcrNdDK2.csv, Energidataservice)					
FcrNd- DK2.csv	PriceTotalEUR where ProductName= "FCR-D upp" AuctionType= "D-1 early"	(1) Pivot long-to-wide: one row per UTC hour, separate columns per product $\times$ auction type combination. (2) Unit conversion: EUR/MW/h $\times 0.746 = \text{øre/kWh}$ (using fixed EUR/DKK = 7.46; 1 EUR/MW = 7.46 DKK/1000 kWh = 0.746 øre/kWh). (3) Negative values clipped to 0 (.clip(lower=0)).	price_ore_kwh_ fcr_d_upp_ d_1_early	$\lambda_t^{\text{FCRD}\uparrow,e}$	øre/kWh
FcrNd- DK2.csv	PriceTotalEUR "FCR-D upp" "D-1 late"	Same pivot and unit conversion as above.	price_ore_kwh_ fcr_d_upp_ d_1_late	$\lambda_t^{\text{FCRD}\uparrow,l}$	øre/kWh
FcrNd- DK2.csv	PriceTotalEUR "FCR-D ned" "D-1 early"	Same pivot and unit conversion as above.	price_ore_kwh_ fcr_d_ned_ d_1_early	$\lambda_t^{\text{FCRD}\downarrow,e}$	øre/kWh

Table 2 continued...

Source file	Raw column(s)	Preprocessing applied	Final CSV column	Symbol	Unit
FcrNd-DK2.csv	PriceTotalEUR "FCR-D ned" "D-1 late"	Same pivot and unit conversion as above.	price_ore_kwh_ fcr_d_ned_ d_1_late	$\lambda_t^{\text{FCRD}\downarrow,l}$	øre/kWh
System frequency (Taajuusdata*.csv, 365 daily files, Fingrid dataset 339)					
Taajuus-data*.csv (365 files)	Value [Hz] sampled at 0.1 s ($\approx 864\,000$ readings/day)	Per-reading activation factor (piecewise linear, Crowley et al. 2025): $a_\tau^\uparrow = \max(0, \min(1, (49.9 - f_\tau)/0.4))$. Hourly mean: $y_t^\uparrow = \frac{1}{n_t} \sum_{\tau \in t} a_\tau^\uparrow$. Hours with $n_t < 18\,000$ readings set <code>data_quality_flag=1</code> and excluded from all model runs.	y_act_fcrd_up	$y_t^{\text{actFCRD}\uparrow}$	$[0, 1]$
Taajuus-data*.csv (365 files)	Value [Hz] sampled at 0.1 s	Per-reading activation factor: $a_\tau^\downarrow = \max(0, \min(1, (f_\tau - 50.1)/0.4))$. Hourly mean as above. Same quality filter applies.	y_act_fcrd_down	$y_t^{\text{actFCRD}\downarrow}$	$[0, 1]$

Derived quantity (not loaded from CSV): The buy-back price $\lambda_t^{\text{FCRDbb}}$ in the late-auction objective is $\max(\lambda_t^{\text{FCRD}\uparrow,e}, \lambda_t^{\text{FCRD}\uparrow,l})$, computed inside `build_late()` in `01_Back_test.ipynb`, and analogously for down.